LSML#4-5

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Lin. model
gradspusk (Sem)

Допустим

X_n $n >> RAM$

One-Hot Encoding

Цвет
кр
син
зел
жел

OHE ⟹

i	цвет_кр	цвет_син	...
0	1		
1	0		
.	.	.	.

Cor цвет марк ~ $x * y$

Hashing TRick

цвет_кр.

① h – хэш функция $X \in R^N \Rightarrow y \in R^M$, $N >> M$

выбираем сами

цвет_кр = 1

$h(\text{"feature"}) = \mathbb{S}\ \in [0 \dots M]$

f_Age = 50

$h(\text{"f_Age"}) = p$

① $h(\text{"}f_1\text{"}) == h(\text{"}f_2\text{"}) \to rand(f_1, f_2) \to$ нестрашно

② $h(\text{"}f_3\text{"}) == h(\text{"}f_4\text{"}) \longrightarrow max(f_3, f_4)$

Если используется OHE, BoW ⟹ пространство признаков разрежено ⟹ Котики будут минимум

или

$max(f_N, f_K) \to 50 \dots$

$= h(\text{"user_id_x"}) = 0$

$h(\text{"f_Age"})$

K=10

$h(\text{"user_id_50"}) \overset{?}{=} h(\text{"f_Age"})$

50 нет

Broadcast

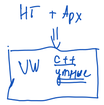
$O(N)$

w. broadcast

New сложно масштабировать

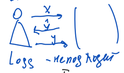
$\sim O(\log N)$

HT + Apx



Online learning

$\mathbb{R} \to$ online



Loss – непрерывный

$Regret = \sum_{i=1}^{T} l_i(w_i) - l_i(w^*)$

$x_t \to \hat{y}_t \to y_t \to w_t$

FTL – follow the leader

$w_{t+1} = \arg\min_w \sum_{i=1}^{t} l_i(w_i)$ → дорого тем
→ всё увеличит

$w_{t+1} = \arg\min_w \sum_{i=1}^{t} l_i(w_i) + r_i(w)$

$r_i = \lambda \cdot \|w - w_i\|_2$

Strong Convexity $F(y) \geq f(x) + \nabla f^T(x)(y-x) + m\|y-x\|_2$

ytd: Lur – SC

$\hat{l}_i(w) = l_i(w_i) + \nabla l_i^T(w_i)(w - w_i)$ – приближаем l в w_i

① $R_{reg} = \sum_{i=1}^{T} \boxed{l_i(w_i)} - l_i(w^*)$

② $R_{reg} = \sum_{i=1}^{T} \boxed{\hat{l}_i(w_i)} - l_i(w^*)$

$\hat{l}_i(w^*) = l_i(w_i) + \nabla l_i^T(w_i)(w^* - w_i)$

$l_i(w^*) \geq l_i(w_i) + \nabla l_i^T(w^* - w_i) + m \cdot \|w^* - w_i\|_2^2$ $m > 0$

w^* – оптимум

$\hat{l}_i(w) = l_i(w_i) + \nabla l_i^T(w_i)(w - w_i)$

SGD

$w_{t+1} = \arg\min_w \sum_{i=1}^{T} l_i(w_i)$

$w_{t+1} = \arg\min_w \sum_{i=1}^{T} \hat{l}_i(w) + r_i(w)$

$\frac{\partial}{\partial w}\Big(\underbrace{l_i(w_i)}_{const} + \underbrace{\nabla l_i^T(w_i)}_{const}(w - w_i) + \lambda\|w - w_i\|_2^2\Big) \to min$

$\nabla l_i(w_i) + 2 \cdot \lambda(w - w_i) = 0$

$2\lambda \cdot (w - w_i) = -\nabla l(w_i)$

$w - w_i = \frac{1}{2\lambda}(-1) \cdot \nabla l_i(w_i)$

$\boxed{w = w_i - \frac{1}{2\lambda}\nabla l_i(w_i)}$ – шаг градиентного спуска

FTRL – proximal

$X \in R^n$ n – большое

$w_{t+1} = \arg\min_w \left(\sum_1^t l_i(w) + r_i(w)\right)$

$w_{t+1} = \arg\min_w \left(\sum_1^t \nabla l_i^T(w_i) \cdot w + \sum_s^t \sigma_s \|w - w_i\| + \lambda\|w\|\right)$

$\nabla l(w_i) = g_i$

$g_{1t} = \sum_1^T g_i$

$w_{t+1} = \arg\min_w \left(g_{1t} \cdot w + \frac{1}{2}\sum_s^T \sigma_s \cdot \|w - w_i\|_2^2 + \lambda\|w\|_2\right)$ $\sum_s^T \sigma_s = \frac{1}{\eta}$

$g_{1t} \cdot w + \frac{1}{2} \cdot \sum_s^t \sigma_s \cdot \|w\|_2^2 + \frac{1}{2} \cdot \sum_{s,s}^T \cancel{\sigma_s \cdot \|w_s\|_2^2} + \sum_s^t 2 \cdot w^T w_s + \lambda\|w\|_2$

$\left(g_{1t} - \sum_1^t \sigma_s w_s\right) w + \frac{1}{2\eta_t} \cdot \|w\|_2 + \lambda\|w\|_2^2 \to min$

$\underbrace{\left(g_{1t} - \sum_1^T \sigma_s w_s\right)}_{Z_0} + \frac{1}{\eta_t} \cdot w + \lambda \frac{\partial}{\partial w}\|w_i\| = 0$

$Z_t = Z_{t-1} + g_t + \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}}\right) \cdot w_t$

$Z_t + \frac{1}{\eta_t} \cdot w + \lambda \underbrace{\frac{\partial}{\partial w}\|w\|_1}_{sgn} = 0$

$w = -\eta_t \cdot (|Z_t| - \lambda)$

посмотрим на норму

$\|Z_t\| < \lambda \Rightarrow w_{t+1} = 0$

$\|Z_t\| \geq \lambda \Rightarrow w_{t+1} = \underbrace{-\eta_t(Z_t - \lambda \cdot sgn(w))}$